Project Plan: BLAST File Transfer over UDP

CSc 412 Computer Networks · Prof. K. Ravindran · **Student:** Md Mamun (individual project) · October 17, 2026

Objective

Implement the BLAST protocol to transfer a 10–15 MB file over UDP with minimal acknowledgment overhead, test it under controlled packet loss, and measure maximum throughput versus loss rate. Platform: macOS (approved by the instructor), C with the UNIX socket API and POSIX threads.

Approach

1. **Packet formats** (`blast.h`): FILE_HDR / FILE_HDR_ACK, DATA with a segment list {(st_rec, end_rec), ...} of up to 16 records, IS_BLAST_OVER(Mst, Mfin), REC_MISS (missing ranges), DISCONNECT / DISCONNECT_ACK.
2. **Phase 1 – setup:** the sender proposes record size (256/512/1024 B), M (200–10000) and the file name; the receiver may lower M to its buffer size. FILE_HDR is retransmitted on timeout.
3. **Phase 2 – blasts:** send M records per blast, then IS_BLAST_OVER; resend only the records listed in REC_MISS (several segments per packet) until REC_MISS is empty. IS_BLAST_OVER is retransmitted on timeout; a sequence number lets the sender ignore stale REC_MISS replies.
4. **Phase 3 – disconnect:** DISCONNECT / DISCONNECT_ACK; the receiver lingers 3 s to answer late retransmissions.
5. **Asynchronous disk I/O:** each side has an M-record buffer B'. A reader thread (sender) and a writer thread (receiver) move data between disk and B' independently of the network, so packets carry a variable number of records, as in Figure 2 of the assignment.
6. **Garbler:** every outgoing packet (data and control, on both sides) is dropped with probability p.

Testing and measurement

- Verify every received file against the original by checksum, at loss rates from 0% to 30%.
- Sweep $p \in \{0, 1, 2, 5, 10, 15, 20, 30\}$%, record size $\in \{256, 512, 1024\}$, M $\in \{500, 2000, 10000\}$, 5 runs each (360 transfers), using a script that records time, throughput, packets sent and dropped, records resent and timeouts.
- Plot throughput versus loss rate for every setting and the maximum achievable throughput.

Schedule

Week	Dates	Work	Status
1	Oct 7–13	Study the specification; basic UDP sender/receiver; packet formats	Done
2	Oct 14–20	Phases 1–3 with timeouts; garbler; asynchronous disk I/O with buffers B'	Done
3	Oct 21–27	Correctness testing under loss; first full measurement run	First run done (Oct 17)
4	Oct 28–Nov 3	Analyze results; explain throughput behaviour (effect of timeouts, M, record size)	Planned
5	Nov 4–10	Possible improvements, e.g. adaptive timeout; rerun measurements if code changes	Planned
6	Nov 11–18	Final report with graphs and source code; submit by Nov 18	Planned

Deliverables

Source code (`sender.c`, `receiver.c`, `blast.h`, `garble.h`, Makefile, experiment and plotting scripts), loss and throughput statistics, a graph of throughput versus packet-loss rate, and a final report. As this is an individual project, all parts are done by Md Mamun.